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Week 4

SESSION-7

Task 1: Process Synchronization and Child Process Monitoring Using waitpid()

To create a child process, synchronize the parent with the child using waitpid(), monitor the child process, check normal termination, retrieve the exit status, handle signal termination, and test process synchronization.

CODE:

```
UW PICO 5.09 File: 21.c Modified
/* Check whether child was stopped */
else if (WIFSTOPPED(status))
{
    printf("Child was stopped by a signal.\n");
    printf("Signal Number: %d\n",
           WSTOPSIG(status));
}

printf("\nParent State: Running\n");
printf("Parent has synchronized with child.\n");
printf("Parent is terminating.\n");
}

return 0;
}
```

⌘ Get Help ⌘ WriteOut ⌘ Read File ⌘ Prev Pg ⌘ Cut Text ⌘ Cur Pos
⌘ Exit ⌘ Justify ⌘ Where is ⌘ Next Pg ⌘ UnCut Text ⌘ To Spell

OUTPUT:

```

[nrk@nallas-mac-Book-Air ~ % gcc 21.c -o 21
[nrk@nallas-mac-Book-Air ~ % ./21
=====
PROCESS SYNCHRONIZATION USING waitpid()
=====
Parent Process
=====
Parent PID : 2436
Child PID : 2437
Parent State: Running

Parent is waiting for child using waitpid()...
Parent State: Waiting

Child Process Started
-----
Child PID : 2437
Parent PID : 2436
Child State: Running

Child is performing some work...
Child is terminating normally.
Child process has finished.

Child terminated normally.
Child Exit Status: 18

Parent State: Running
Parent has synchronized with child.
Parent is terminating.
nrk@nallas-mac-Book-Air ~ % █

```

Task 2: PATH Variable Retrieval and Command Resolution

To retrieve the PATH variable, parse its search directories, locate executable files, verify execute permissions, handle missing commands, and test command resolution.

CODE:

```

UW PICO 5.09                                     File: 22.c                               Modified
count++;
    directory = strtok(NULL, ":");
}
    free(copy);
}

/* Get command from user */
printf("\nEnter command to resolve: ");
scanf("%255s", command);

/* Resolve command */
resolve_command(command);

return 0;
}

```

OUTPUT:

```
env/global/bin:/opt/homebrew/bin:/usr/local/mysql/bin:/usr/local/mysql/bin:/usr/local/mysql/bin:/usr/local/mysql/bin:/usr/local/mysql/bin

PATH Search Directories:
-----
1. /Library/Frameworks/Python.framework/Versions/3.14/bin
2. /Library/Frameworks/Python.framework/Versions/3.14/bin
3. /usr/local/bin
4. /System/Cryptexes/App/usr/bin
5. /usr/bin
6. /bin
7. /usr/sbin
8. /sbin
9. /var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin
10. /var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/bin
11. /var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin
12. /pkg/env/global/bin
13. /opt/homebrew/bin
14. /usr/local/mysql/bin
15. /usr/local/mysql/bin
16. /usr/local/mysql/bin
17. /usr/local/mysql/bin
18. /usr/local/mysql/bin

Enter command to resolve: 4
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/4
Checking: /Library/Frameworks/Python.framework/Versions/3.14/bin/4
Checking: /usr/local/bin/4
Checking: /System/Cryptexes/App/usr/bin/4
Checking: /usr/bin/4
Checking: /bin/4
Checking: /usr/sbin/4
Checking: /sbin/4
Checking: /var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin/4
Checking: /var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/bin/4
Checking: /var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin/4
Checking: /pkg/env/global/bin/4
Checking: /opt/homebrew/bin/4
Checking: /usr/local/mysql/bin/4
Checking: /usr/local/mysql/bin/4
Checking: /usr/local/mysql/bin/4
Checking: /usr/local/mysql/bin/4
Checking: /usr/local/mysql/bin/4

=====
COMMAND NOT FOUND
=====
Command '4' was not found in PATH.
nrk@nallas-mac-Book-Air ~ % ls
1.c.save          2          5.c              fed.css          project.html
11                2.c        a.c.save         file.c           Public
11.c              20         airline-lost-found file.c           public class SegmentTreeLazy { .java
13                20.c       ancient dress.fbx import java.util.java
13.C              21         Applications     kl.c             Rohith
14                21.c       b.c.save        Library          Rohith.c
14.c              22         blenderkit_data lk               space.save
15                22.c       class AVLTree (.java lk.c            student.html.h
15.c              3          class RankNode (.java logism.circ     student registration form.html
16                3.c        DaVinci Resolve Media Movies           student.html.h
16.c              4          ddca.circ       Music            Untitled-1.cs
17                4.c        Desktop         n.c             .
17.c              444        Documents       New Folder
19                444.c      Downloads      package com.mediflow.java
19.c              5          EXAMINATION SEATING ARRANGEMENT Pictures

nrk@nallas-mac-Book-Air ~ %
```

SESSION-8

Task 1: Variable Reference Detection and Value Expansion

To detect variable references, expand environment variable values, handle undefined variables, update tokens after expansion, support nested variables, and test expansion logic.

CODE:

```
UW PICO 5.09                                     File: 23.c                                     Modified
printf("          EXPANSION RESULT\n");
printf("===== \n");
printf("Original : %s\n", input);
printf("Expanded : %s\n", expanded);
printf("\nExpansion completed successfully.\n");
return 0;
}
```

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OUTPUT:

```

nrk@nallas-mac-Book-Air ~ %
nrk@nallas-mac-Book-Air ~ % gcc 23.c -o 23
nrk@nallas-mac-Book-Air ~ % ./23
=====
VARIABLE EXPANSION PROGRAM
=====

Examples:
$HOME
$USER
${PATH}
User: $USER, Home: $HOME
Undefined: $ABXYZ

Enter input: pwd

Detected Variables:
=====
No variable references found.
=====
EXPANSION RESULT
=====
Original : pwd
Expanded : pwd

Expansion completed successfully.
nrk@nallas-mac-Book-Air ~ % █

```

Task 2: Built-in Identification, Dispatch Table and In-Process Execution

To identify built-in commands, create a dispatch table, execute in-process commands, handle invalid commands, maintain shell state, and test dispatch logic.

CODE:

```

UW PICO 5.09 File: 24.c Modified
{
    /*
     * Execute built-in in the current process.
     * No fork() is used.
     */
    state.last_status =
        command_function(args, &state);
}
else
{
    printf("myshell: command not found: %s\n", args[0]);
    state.last_status = 127;
}
}

printf("Shell terminated.\n");
return state.last_status;
}█

```

Output:

```
Last login: Sat Aug 29 09:22:42 on ttys000
nrk@nallas-mac-Book-Air ~ % nano 24.c
```

```
nrk@nallas-mac-Book-Air ~ % gcc 24.c -o 24
nrk@nallas-mac-Book-Air ~ % ./24
=====
BUILT-IN COMMAND SHELL
=====
Type 'help' to see available commands.
Type 'exit' to quit.
myshell> █
```